

Practical 3

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `rows` and `columns`, representing the number of rows and the number of columns.
- The next `rows` lines contain `columns` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

```
CODE: import numpy as np
rows, cols = map(int, input().split())

elements = []
for _ in range(rows):
    elements.extend(map(int, input().split()))

array = np.array(elements).reshape(rows, cols)

print(array)

print(array.ndim)

print(array.shape)

print(array.size)
```


3.2.1. Numpy: Matrix Operations

The given code takes two matrices, A and B , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition ()
2. Subtraction ()
3. Element-wise Multiplication ()
4. Matrix Multiplication ()
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for A , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for B , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

CODE:

```
import numpy as np
```

```

# Input matrices print("Enter Matrix A:") matrix_a =
np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:") matrix_b = np.array([list(map(int,
input().split())) for i in range(3)])

# Addition addition_result = matrix_a +
matrix_b print("Addition (A + B):")
print(addition_result) # Subtraction
subtraction_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(subtraction_result)

# Multiplication (element-wise) elementwise_multiplication_result
= matrix_a * matrix_b print("Element-wise Multiplication (A *
B):") print(elementwise_multiplication_result)

# Matrix multiplication (dot product)

matrix_multiplication_result = np.dot(matrix_a, matrix_b) print("A
dot B:")

print(matrix_multiplication_result)

# Transpose transpose_a
= matrix_a.T
print("Transpose of A:")
print(transpose_a)

```

matrixOp... Submit

Average time
0.057 s
56.75 ms

Maximum time
0.109 s
109.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 100 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A + B):	Addition (A + B):
[[2 3 4]]	[[2 3 4]]
[5 6 7]	[5 6 7]
[8 9 10]]	[8 9 10]]
Subtraction (A - B):	Subtraction (A - B):
[[0 1 2]]	[[0 1 2]]
[3 4 5]	[3 4 5]
[6 7 8]]	[6 7 8]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[1 2 3]]	[[1 2 3]]
[4 5 6]	[4 5 6]
[7 8 9]]	[7 8 9]]
A dot B:	A dot B:
[[6 6 6]]	[[6 6 6]]
[15 15 15]	[15 15 15]
[24 24 24]]	[24 24 24]]
Transpose of A:	Transpose of A:
[[1 4 7]]	[[1 4 7]]
[2 5 8]	[2 5 8]
[3 6 9]]	[3 6 9]]

Test case 2 43 ms Debug	
Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
2 4 8	2 4 8
9 6 5	9 6 5
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
10 12 13	10 12 13
14 15 16	14 15 16
17 18 19	17 18 19
Addition (A + B):	Addition (A + B):
[[12 16 21]]	[[12 16 21]]
· [23 21 21]	· [23 21 21]
· [24 26 28]]	· [24 26 28]]
Subtraction (A - B):	Subtraction (A - B):
[[-8 -8 -5]]	[[-8 -8 -5]]
· [-5 -9 -11]	· [-5 -9 -11]
· [-10 -10 -10]]	· [-10 -10 -10]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[-20 -48 104]]	[[-20 -48 104]]
· [126 -90 -80]	· [126 -90 -80]
· [119 144 171]]	· [119 144 171]]
A dot B:	A dot B:
[[212 228 242]]	[[212 228 242]]
· [259 288 308]	· [259 288 308]
· [335 366 390]]	· [335 366 390]]
Transpose of A:	Transpose of A:
[[2 9 7]]	[[2 9 7]]
· [4 6 8]	· [4 6 8]
· [8 5 9]]	· [8 5 9]]

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

08:24

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).
- Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np
```

```
# Input matrices print("Enter Array1:") arr1 =  
np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Array2:") arr2 = np.array([list(map(int,  
input().split())) for i in range(3)])
```

```
# Perform horizontal stacking (hstack)  
a=np.hstack((arr1,arr2))  
print("Horizontal Stack:") print(a)
```

```
# Perform vertical stacking (vstack)  
b=np.vstack((arr1,arr2))  
print("Vertical Stack:") print(b)
```


Average time
0.052 s
52.00 ms

Maximum time
0.066 s
66.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 60 ms Debug

Expected output

Actual output

Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[1 2 3 4 5 6]
[4 5 6 7 8 9]
[7 8 9 10 11 12]]
Vertical Stack:
[[1 2 3]
[4 5 6]
[7 8 9]
[4 5 6]
[7 8 9]
[10 11 12]]

Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[1 2 3 4 5 6]
[4 5 6 7 8 9]
[7 8 9 10 11 12]]
Vertical Stack:
[[1 2 3]
[4 5 6]
[7 8 9]
[4 5 6]
[7 8 9]
[10 11 12]]

Test case 2 45 ms Debug

Expected output

Actual output

Enter Array1:
-1 -2 -3
-4 -5 -6
-7 -8 -9
Enter Array2:
10 11 12
13 14 15
16 17 18
Horizontal Stack:
[[-1 -2 -3 10 11 12]
[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]
Vertical Stack:
[[-1 -2 -3]
[-4 -5 -6]
[-7 -8 -9]
[10 11 12]
[13 14 15]
[16 17 18]]

Enter Array1:
-1 -2 -3
-4 -5 -6
-7 -8 -9
Enter Array2:
10 11 12
13 14 15
16 17 18
Horizontal Stack:
[[-1 -2 -3 10 11 12]
[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]
Vertical Stack:
[[-1 -2 -3]
[-4 -5 -6]
[-7 -8 -9]
[10 11 12]
[13 14 15]
[16 17 18]]

3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

```
import numpy as np
```

```
# Take user input for the start, stop, and step of the sequence
```

```
start = int(input()) stop = int(input()) step = int(input())
```

```
# Generate the sequence using np.arange() n=np.arange(start,stop,  
step)
```

```
# Print the generated sequence print(n)
```

Average time
0.058 s
57.75 ms

Maximum time
0.146 s
146.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 146 ms Debug

Expected output	Actual output
3	3
18	18
2	2
[3 5 7 9]	[3 5 7 9]

Test case 2 29 ms Debug

Expected output	Actual output
18	18
28	28
38	38
[18]	[18]

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

08:30

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np
```

```
def array_operations(A, B):
```

```
    # Convert A and B to NumPy arrays
```

```
    A = np.array(A)
```

```
    B = np.array(B) # Arithmetic Operations
```

```
    sum_result = A + B diff_result = A - B
```

```
    prod_result = A * B
```

```
    # Statistical Operations
```

```
    mean_A = np.mean(A)
```

```
    median_A = np.median(A)
```

```
    std_dev_A = np.std(A)
```

```
    # Bitwise Operations
```

```
    and_result = A & B or_result =
```

```
    A | B xor_result = A ^ B
```

```
    # Output results with one space between each element    print("Element-
wise Sum:", ' '.join(map(str, sum_result)))    print("Element-wise
Difference:", ' '.join(map(str, diff_result)))    print("Element-wise Product:", '
'.join(map(str, prod_result)))
```

```

print(f'Mean of A: {mean_A}')

print(f'Median of A: {median_A}') print(f'Standard
Deviation of A: {std_dev_A}')

print("Bitwise AND:", ' '.join(map(str, and_result)))

print("Bitwise OR:", ' '.join(map(str, or_result)))    print("Bitwise
XOR:", ' '.join(map(str, xor_result)))

```

```

A = list(map(int, input().split())) # Elements of array A B
= list(map(int, input().split())) # Elements of array B
array_operations(A, B)

```

Average time: **0.065 s** 65.33 ms Maximum time: **0.143 s** 143.00 ms

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 143 ms Debug

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5. Numpy: Copying and Viewing Arrays

04:23

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array. □ Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array original_array =  
np.array(inputlist)
```

```
# Create a view
```

```
view_array =original_array.view()
```

```
# Create a copy copy_array
```

```
=original_array.copy()
```

```
# Modify the view view_array[0] = 99 print("Original array  
after modifying view:", original_array) print("View array:",  
view_array)
```

```
# Modify the copy copy_array[1] = 88 print("Original array  
after modifying copy:", original_array) print("Copy array:",  
copy_array)
```

Average time: 0.018 s (18.50 ms) | Maximum time: 0.032 s (32.00 ms) | 2 out of 2 shown test case(s) passed | 2 out of 2 hidden test case(s) passed

Test case 1 (32 ms) [Debug] [Test]

Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]	Copy array: [10 88 30 40 50 60 70 80]

Test case 2 (17 ms) [Debug] [Test]

Expected output	Actual output
99 2 3 4 5	99 2 3 4 5
Original array after modifying view: [99 2 3 4 5]	Original array after modifying view: [99 2 3 4 5]
View array: [99 2 3 4 5]	View array: [99 2 3 4 5]
Original array after modifying copy: [99 2 3 4 5]	Original array after modifying copy: [99 2 3 4 5]
Copy array: [99 88 3 4 5]	Copy array: [99 88 3 4 5]

3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

05:03

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where `search_value` appears in `array1` and print these indices.
2. Counting: Count how many times `count_value` appears in `array1` and print the count.
3. Broadcasting: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
4. Sorting: Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.
4. The sorted array.

```
import numpy as np
```

```
# Input array from the user array1 =  
np.array(list(map(int, input().split())))
```



```

# Searching search_value = int(input("Value
to search: ")) count_value = int(input("Value
to count: ")) broadcast_value =
int(input("Value to add: "))

# Find indices where value matches in array1
a=np.where(array1==search_value)[0] print(a)

# Count occurrences in array1 b=np.count_nonzero(array1==count_value)
print(b) # Broadcasting
addition c=array1 +
broadcast_value print(c) #
Sort the first array
d=np.sort(array1) print(d)

```

Average time: 0.035 s (34.75 ms) Maximum time: 0.061 s (61.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 81 ms Debug

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
Value to search: 1	Value to search: 1
Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3
[3 3 3 4 4 4]	[3 3 3 4 4 4]
[1 1 1 2 2 2]	[1 1 1 2 2 2]

Test case 2 27 ms Debug

Expected output	Actual output
1 2 3 4 5	1 2 3 4 5
Value to search: 6	Value to search: 6
Value to count: 6	Value to count: 6
Value to add: 6	Value to add: 6
[]	[]
0	0
[-7 -8 -9 -10 -11]	[-7 -8 -9 -10 -11]
[1 2 3 4 5]	[1 2 3 4 5]

3.2.7. Student Data Analysis and Operations

31:34

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students: Determine the total number of students in the dataset.
- Print all student roll numbers: Extract and print the roll numbers of all students.
- Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.
- Find the average marks of each student: Compute the average marks for each student.
- Find average marks of each subject: Compute the average marks for all students in each subject.

- Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Find the count of students who got less than 40 marks in Subject 1: Count the number of students who scored less than 40 marks in Subject 1.
- Find the count of students who got more than 90 marks in Subject 2: Count the number of students who scored more than 90 marks in Subject 2.
- Find the count of students who scored ≥ 90 in each subject: Count the number of students who scored 90 or more marks in each subject.
- Find the count of subjects in which each student scored ≥ 90 : Determine how many subjects each student scored 90 or more marks in.
- Print Subject 1 marks in ascending order: Sort and print the marks of students in Subject 1 in ascending order.
- Print students who scored between 50 and 90 in Subject 1: Display students who scored marks between 50 and 90 in Subject 1.
- Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

CODE:

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

1. Print all student details print("All
student Details:\n", a)

2. print total students

print("Total Students:",len(a))

3. Print all student Roll numbers print("All
Student Roll Nos", a[:,0])

4. Print subject 1 marks
print("Subject 1 Marks", a[:,1])

5. print minimum marks of Subject 2 print("Min
marks in Subject 2", np.min(a[:, 2]))

6. print maximum marks of Subject 3 print("Max
marks in Subject 3", np.max(a[:,3]))

7. Print All subject marks print("All
subject marks:", a[:,1:])

8. print Total marks of students print("Total
Marks", np.sum(a[:,1:], axis = 1))

9. print average marks of each student print(np.round(np.mean(a[:,1:],
axis = 1),1))

```
# 10. print average marks of each subject print("Average Marks of  
each subject", np.mean(a[:, 1:], axis = 0) )
```

```
# 11. print average marks of S1 and S2 print("Average Marks of S1  
and S2", np.mean(a[:, 1:3],axis=0) )
```

```
# 12. print average marks of S1 and S3 print("Average Marks of S1 and S3",  
np.mean(a[:, [1, 3]], axis = 0) )
```

```
# 13. print Roll number who got maximum marks in Subject 3 max_sub3_index =  
np.argmax(a[:, 3]) print("Roll no who got maximum marks in Subject 3",  
a[max_sub3_index , 0] ) # 14. print Roll number who got minimum marks in  
Subject 2 min_sub2_index = np.argmin(a[:, 2]) print("Roll no who got minimum  
marks in Subject 2", a[min_sub2_index, 0] )
```

```
# 15. print Roll number who got 24 marks in Subject 2
```

```
print(f"Roll no who got 24 marks in Subject 2 [{a[:,2] == 24}[:,0]]" )
```

```
# 16. print count of students who got marks in Subject 1 < 40 count_sub1_lt_40 =  
np.sum(a[:,1]<40) print("Count of students who got marks in Subject 1 < 40",  
count_sub1_lt_40 )
```

```
# 17. print count of students who got marks in Subject 2 > 90 count__sub2_gt_90 =  
np.sum(a[:,2]>90) print("Count of students who got marks in Subject 2 > 90:",  
count__sub2_gt_90 )
```

```
# 18. print count of students in each subject who got marks >= 90 count_each_subject_90plus  
= np.sum(a[:,1:]>=90 , axis = 0)
```

```
print("Count of students in each subject who got marks >= 90:", count_each_subject_90plus
)
```

```
# 19. print count of subjects in which each student got marks >= 90 print("Roll
no:", a[:,0].astype(float)    )
print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:] >= 90 , axis = 1)
)
```

```
# 20. Print S1 marks in ascending order print(np.sort(a[:,1]))
```

```
# 21. Print S1 marks >= 50 and <= 90
s1_filteredv = a[(a[:,1]>=50) & (a[:,1]<=90)]
print(s1_filteredv) print(a)
```

```
# 22. Print the index position of marks 79
```

```
indices_79 = np.where(a[:,1:]==79)[0] print((indices_79,))
```

Average time
0.103 s
103.00 ms

Maximum time
0.103 s
103.00 ms

1 out of 1 shown test case(s) passed

Test case 1 103 ms Debug

Expected output

Actual output

All student Details:	All student Details:
[301, 67, 77, 88]	[301, 67, 77, 88]
[302, 78, 88, 77]	[302, 78, 88, 77]
[303, 45, 56, 89]	[303, 45, 56, 89]
[304, 88, 98, 45]	[304, 88, 98, 45]
[305, 78, 88, 99]	[305, 78, 88, 99]
[306, 68, 78, 36]	[306, 68, 78, 36]
[307, 79, 12, 45]	[307, 79, 12, 45]
[308, 46, 45, 77]	[308, 46, 45, 77]
[309, 89, 32, 88]	[309, 89, 32, 88]
[310, 79, 24, 33]	[310, 79, 24, 33]
Total Students: 10	Total Students: 10
All Student Roll Nos: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]	All Student Roll Nos: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]
Subject 1 Marks: [67, 78, 45, 88, 78, 68, 79, 46, 89, 79]	Subject 1 Marks: [67, 78, 45, 88, 78, 68, 79, 46, 89, 79]
Min marks in Subject 2: 12.0	Min marks in Subject 2: 12.0
Max marks in Subject 3: 99.0	Max marks in Subject 3: 99.0
All subject marks: [[67, 77, 88],	All subject marks: [[67, 77, 88],
[78, 88, 77],	[78, 88, 77],
[45, 56, 89],	[45, 56, 89],
[88, 98, 45],	[88, 98, 45],
[78, 88, 99],	[78, 88, 99],
[68, 78, 36],	[68, 78, 36],
[79, 12, 45],	[79, 12, 45],
[46, 45, 77],	[46, 45, 77],
[89, 32, 88]]	[89, 32, 88]]

2025

Dehinner